

Numerical models (3/3): EX 11–15

Outline

EX 11 — Shoes with a minimum production threshold

EX 12 — Balancing between two workers

EX 13 — Emergency department shifts

EX 14 — Shoes: production, inventory and hirings

EX 15 — The music school timetable

The five parts of every numerical model

Every numerical model always has the same five parts:

- ▶ the statement, with the data of the instance;
- ▶ the **variables**, with their domain and their count;
- ▶ the **model on the instance**, primal and dual;
- ▶ a feasible solution built by hand, which gives the primal bound;
- ▶ a dual solution built by hand, which gives the dual bound, and the comparison with the solver's optimum.

EX 11 — Shoes with a minimum production threshold

Introduction: EX 11

- ▶ A shoe company chooses which types of shoe to produce and how many pairs
- ▶ If a type is produced, at least a **minimum threshold** of it is produced
- ▶ Techniques: minimum thresholds and **activations** with a big-M read off the data
- ▶ We will see a case in which primal and dual bound coincide immediately

EX 11 — three types of shoe with a threshold

A shoe company produces hiking boots, loafers and walking shoes. The table reports the resources needed for one pair, the minimum quantity to be produced if the type is produced, and the selling price:

Type	leather (g)	hours	nails	minimum	price
hiking boots	850	3	20	100	150
loafers	600	2	15	200	120
walking shoes	700	2.5	20	150	130

There are 120 000 grams of leather, 7000 machine hours and 40 000 nails available. The goal is to maximise the revenue.

The variables

Two families of variables, one per type of shoe:

- ▶ x_j : the pairs produced of type j , integer and non-negative;
 - ▶ y_j : worth 1 if type j is produced and 0 otherwise.
-
- ▶ It is again model 9.3 without the bonus, with three resources
 - ▶ The big-Ms read off the data are $M = (141, 200, 171)$, all limited by the leather
 - ▶ For the loafers threshold and maximum coincide (200): the variable x_2 is in fact a binary multiplied by 200

EX 11 on the instance — primal

EX 11 on the instance — primal

$$\begin{array}{llll} \max & 150x_1 + 120x_2 + 130x_3 & & \\ \text{subject to} & 850x_1 + 600x_2 + 700x_3 & & \leq 120000 \\ & 3x_1 + 2x_2 + \frac{5}{2}x_3 & & \leq 7000 \\ & 20x_1 + 15x_2 + 20x_3 & & \leq 40000 \\ & x_1 & -100y_1 & \geq 0 \\ & & x_2 & -200y_2 \geq 0 \\ & & & x_3 -150y_3 \geq 0 \\ & x_1 & -141y_1 & \leq 0 \\ & & x_2 & -200y_2 \leq 0 \\ & & & x_3 -171y_3 \leq 0 \\ & x_1, & x_2, & x_3 \in \mathbb{Z}_{\geq 0} \\ & & & y_1, \quad y_2, \quad y_3 \in \{0, 1\} \end{array}$$

A feasible solution: the primal bound

- ▶ Constructive heuristic on the price per gram of leather: 3/17 for the boots, 1/5 for the loafers, 13/70 for the walking shoes
- ▶ One starts from the loafers, produces 200 of them and the leather runs out exactly
- ▶ No other type reaches its own threshold

$$z(MILP) \geq LB = 24\,000.$$

A dual solution: the dual bound

With $\ell = \beta = 0$ and a single resource priced:

leather: $\frac{1}{5} \Rightarrow 24\,000$

hours: $60 \Rightarrow 420\,000$

nails: $8 \Rightarrow 320\,000$

The better bound is the one from the leather:

$$z(MILP) \leq UB = 24\,000.$$

The final comparison

	LB	UB	$z(LP)$	$z(LP^+)$	$z(MILP)$
EX 11	24 000	24 000	24 000	24 000	24 000

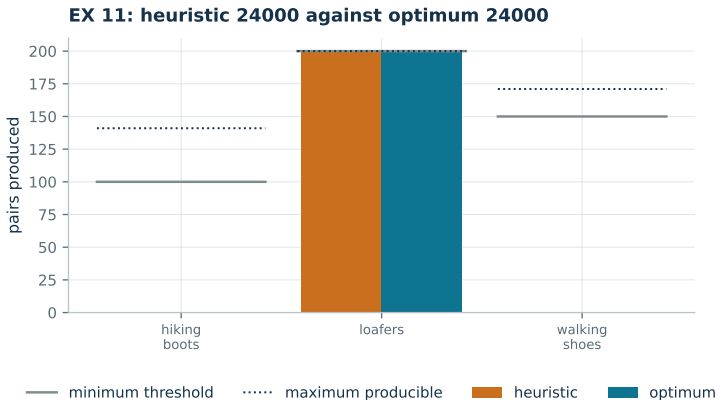
- ▶ All the numbers coincide: the sandwich closes immediately
- ▶ A single resource is tight, the leather, and the type that gets the most out of it per gram uses it up on its own, within its own threshold and maximum
- ▶ When the dual bound matches a feasible solution, the problem is over

Two variations

- ▶ **Doubled thresholds:** the one for the loafers becomes 400 pairs while the leather allows at most 200. No type reaches its own threshold any more and production stops altogether:
 $z(MILP) = 0$
- ▶ **Leather at 200 000 grams:** 333 loafers are produced for 39 960 euros, and the big-Ms must be recomputed: they become (235, 333, 285)

Big-M values are not constants of the model,
but functions of the data of the instance.

EX 11: heuristic and optimum



The pairs produced by the heuristic and the optimal ones, with the minimum threshold and the maximum producible of each type.

Summary

- ▶ Minimum threshold: $x_j \geq m_j y_j$; activation: $x_j \leq M_j y_j$
- ▶ The big-M is read off the data: it is the maximum producible with the tightest resource
- ▶ Heuristic on the price per gram of leather: $LB = 24\,000$
- ▶ Dual with only the leather priced: $UB = 24\,000$
- ▶ The two bounds coincide: optimality is proved by hand

EX 12 — Balancing between two workers

Introduction: EX 12

- ▶ Four jobs to be split between two workers
- ▶ “Balanced” is not yet an objective: it must be translated linearly
- ▶ Technique: **min-max** with an auxiliary variable z
- ▶ We will build the dual by hand and see that the two bounds meet

EX 12 — workloads as balanced as possible

Four indivisible jobs of duration 2, 3, 6 and 7 must be assigned to two workers so that their workloads are as balanced as possible.

Observation 1: “Balanced” must be translated, and the translation declared

“Balanced workloads” is not an objective: it is a word. The two linear translations: minimise the maximum of the loads, or minimise their difference. Since the total $q = 18$ is constant,

$$\max(w_1, w_2) = \frac{q}{2} + \frac{|w_1 - w_2|}{2},$$

so the two models have the same optimal solutions — but different optimal values: 9 for the first, 0 for the second. Reporting “9” as the “difference between the loads” is a mistake, and so is reporting “0” as the “maximum load”.

The variables

- ▶ The data are $n = 4$ jobs of duration d_j
- ▶ The total load is $q = \sum_{j=1}^n d_j = 18$
- ▶ $x_j = 1$ if job j goes to worker 1, 0 if it goes to worker 2
- ▶ $z \geq 0$ is the maximum load

EX 12 on the instance — primal

EX 12 on the instance — primal

$$\begin{array}{ll}\min & z \\ \text{subject to} & 2x_1 + 3x_2 + 6x_3 + 7x_4 - z \leq 0 \\ & -2x_1 - 3x_2 - 6x_3 - 7x_4 - z \leq -18 \\ & x_1, \quad x_2, \quad x_3, \quad x_4 \in \{0, 1\} \\ & z \geq 0\end{array}$$

A feasible solution: the primal bound

LPT: the jobs in decreasing order of duration, each to the least loaded worker.

- ▶ Job 4 (7) to worker 1
- ▶ Job 3 (6) to worker 2
- ▶ Job 2 (3) to worker 2, which reaches 9
- ▶ Job 1 (2) to worker 1, which reaches 9

Loads (9, 9), so $UB = 9$.

EX 12 on the instance — dual

EX 12 on the instance — dual

$$\begin{array}{ll}\max & -18\pi_2 \\ \text{subject to} & 2\pi_1 - 2\pi_2 \leq 0 \\ & 3\pi_1 - 3\pi_2 \leq 0 \\ & 6\pi_1 - 6\pi_2 \leq 0 \\ & 7\pi_1 - 7\pi_2 \leq 0 \\ & -\pi_1 - \pi_2 \leq 1 \\ & \pi_1, \quad \pi_2 \leq 0\end{array}$$

Description of the dual

- ▶ π_1 and π_2 are the (non-positive) prices of the two min-max constraints
- ▶ The objective prices the total load q at π_2
- ▶ The first constraints are the columns of the x_j : moving job j from one worker to the other shifts both loads by d_j , and the balance cannot be positive because the x_j do not appear in the objective
- ▶ The last constraint is the column of z : coefficient -1 in both constraints, cost 1 in the primal objective

A dual solution: the dual bound

- ▶ Since $d_j > 0$, the first constraints say $\pi_1 \leq \pi_2$
- ▶ Setting $\pi_1 = \pi_2 = t$, the constraint from z gives $-2t \leq 1$, that is $t \geq -1/2$
- ▶ The objective $-qt$ is largest at $t = -1/2$:

$$LB = \frac{q}{2} = 9.$$

Meaning: “the two loads add up to q ,
so the larger one is at least $q/2$ ”.

The same thing with non-negative dual variables

- ▶ Many treatments prefer dual variables ≥ 0
- ▶ It suffices to set $\alpha = -\pi_1$ and $\beta = -\pi_2$: the dual becomes

$$\max q\beta \quad \text{with} \quad \alpha \leq \beta, \quad \alpha + \beta \leq 1$$

- ▶ $\alpha = \beta = 1/2$ gives 9 again
- ▶ It is the *same* solution written with the other sign
- ▶ What one cannot do is declare variables ≥ 0 and then use one of them negative

The final comparison

$z(MILP) = 9$ with jobs $\{1, 4\}$ to the first worker and $\{2, 3\}$ to the second: loads $(9, 9)$, perfectly balanced.

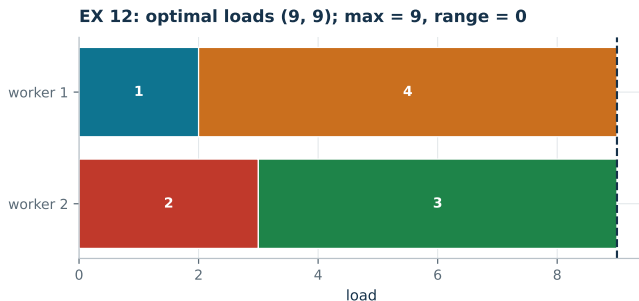
UB (LPT)	LB (dual)	$z(LP)$	$z(MILP)$	gap
9	9	9	9	0.0%

- ▶ The two hand-built bounds meet: optimality is proved without solving the MILP
- ▶ This happens because q is even and the jobs can be split exactly in half

What if q were odd?

- ▶ The dual of the relaxation would still give $q/2$, which is not an integer
- ▶ It is the integrality of the loads that allows strengthening it to $\lceil q/2 \rceil$
- ▶ That is a combinatorial argument, not a dual bound

EX 12: the optimal loads



The optimal loads. The dashed line is $q/2$, the dual bound.

Summary

- ▶ “Balanced” must be translated: min-max or difference, with different optimal values (9 and 0)
- ▶ Min-max: $z \geq w_1, z \geq w_2, \min z$
- ▶ LPT gives $UB = 9$; the dual with $\pi_1 = \pi_2 = -1/2$ gives $LB = 9$
- ▶ Dual variables ≤ 0 or ≥ 0 : the same solution, as long as one is consistent with the declared sign

EX 13 — Emergency department shifts

Introduction: EX 13

- ▶ Covering the daily requirement of an emergency department with fixed shift patterns
- ▶ It is a **set covering** with coefficients 1 and $1/2$ and integer, non-binary variables
- ▶ Primal bound with a covering heuristic, dual bound with a uniform price
- ▶ We will find that the part-time day does not pay off, and only the model shows it

EX 13 — weekly shifts (1/2)

In an emergency department the number of full-time staff members required is given for every day of the week:

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
staff	10	8	12	9	9	7	8

One full-time member can be replaced by two part-time members.

EX 13 — weekly shifts (2/2)

- ▶ The shift pattern is fixed: four consecutive full-time days, then one part-time day, then two rest days
- ▶ The weekly cost of a worker depends on the days worked: 100 euros for every weekday (Monday to Friday), 110 for Saturday and 130 for Sunday
- ▶ These costs are halved on part-time days
- ▶ The goal is the weekly plan of minimum cost

The variables

- ▶ $n = 7$ shift patterns, one per starting day
- ▶ $m = 7$ days to cover, with requirement b_i
- ▶ x_j : number of workers following pattern j (integer variables)
- ▶ c_j : weekly cost of pattern j
- ▶ a_{ij} : share of day i covered by pattern j (1 on the four full days, 1/2 on the part-time day, 0 on the two rest days)

EX 13 on the instance — primal

EX 13 on the instance — primal

$$\min 450x_1 + 455x_2 + 475x_3 + 490x_4 + 490x_5 + 490x_6 + 480x_7$$

$$\begin{array}{llllllll} \text{subject to} & x_1 & & & +\frac{1}{2}x_4 & +x_5 & +x_6 & +x_7 \geq 10 \\ & x_1 & +x_2 & & & +\frac{1}{2}x_5 & +x_6 & +x_7 \geq 8 \\ & x_1 & +x_2 & +x_3 & & & +\frac{1}{2}x_6 & +x_7 \geq 12 \\ & x_1 & +x_2 & +x_3 & +x_4 & & & +\frac{1}{2}x_7 \geq 9 \\ & \frac{1}{2}x_1 & +x_2 & +x_3 & +x_4 & +x_5 & & \geq 9 \\ & & \frac{1}{2}x_2 & +x_3 & +x_4 & +x_5 & +x_6 & \geq 7 \\ & & & \frac{1}{2}x_3 & +x_4 & +x_5 & +x_6 & +x_7 \geq 8 \\ & x_1, & x_2, & x_3, & x_4, & x_5, & x_6, & x_7 \in \mathbb{Z}_{\geq 0} \end{array}$$

The costs of the patterns

Pattern	full days	part-time	cost
1 (starts Mon)	Mon, Tue, Wed, Thu	Fri	450
2 (starts Tue)	Tue, Wed, Thu, Fri	Sat	455
3 (starts Wed)	Wed, Thu, Fri, Sat	Sun	475
4 (starts Thu)	Thu, Fri, Sat, Sun	Mon	490
5 (starts Fri)	Fri, Sat, Sun, Mon	Tue	490
6 (starts Sat)	Sat, Sun, Mon, Tue	Wed	490
7 (starts Sun)	Sun, Mon, Tue, Wed	Thu	480

For instance pattern 3 costs $100 + 100 + 100 + 110$ for the four full days (Wed, Thu, Fri, Sat) plus $130/2 = 65$ for the part-time Sunday, that is 475 in total.

A feasible solution: the primal bound

Constructive covering heuristic: while some demand is uncovered one adds a copy of the pattern with the lowest ratio

$$\frac{\text{cost}}{\text{demand actually covered}},$$

recomputed at every step.

- ▶ On the instance 18 shifts are added: eight of pattern 1 (at 100 euros per unit covered), then four of pattern 3, two of pattern 5, one of pattern 4 and three of pattern 7
- ▶ The ratios steadily worsen, up to 480 for the last unit

$$z(MILP) \leq UB = 8410.$$

EX 13 on the instance — dual

EX 13 on the instance — dual

$$\begin{aligned} & \max 10\pi_1 + 8\pi_2 + 12\pi_3 + 9\pi_4 + 9\pi_5 + 7\pi_6 + 8\pi_7 \\ \text{subject to} \quad & \pi_1 + \pi_2 + \pi_3 + \pi_4 + \frac{1}{2}\pi_5 \leq 450 \\ & \pi_2 + \pi_3 + \pi_4 + \pi_5 + \frac{1}{2}\pi_6 \leq 455 \\ & \pi_3 + \pi_4 + \pi_5 + \pi_6 + \frac{1}{2}\pi_7 \leq 475 \\ & \frac{1}{2}\pi_1 + \pi_4 + \pi_5 + \pi_6 + \pi_7 \leq 490 \\ & \pi_1 + \frac{1}{2}\pi_2 + \pi_5 + \pi_6 + \pi_7 \leq 490 \\ & \pi_1 + \pi_2 + \frac{1}{2}\pi_3 + \pi_6 + \pi_7 \leq 490 \\ & \pi_1 + \pi_2 + \pi_3 + \frac{1}{2}\pi_4 + \pi_7 \leq 480 \\ & \pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6, \pi_7 \geq 0 \end{aligned}$$

A dual solution: the dual bound

The *best ratio*: the same price t on all the days. Every pattern covers $4 + 1/2 = 9/2$ person-days, so $\frac{9}{2} t \leq c_j$ and

$$t = \min_j \frac{c_j}{9/2} = \min\left(100, \frac{910}{9}, \frac{950}{9}, \frac{980}{9}, \frac{980}{9}, \frac{980}{9}, \frac{320}{3}\right) = 100,$$

attained by pattern 1. The total demand is $10 + 8 + 12 + 9 + 9 + 7 + 8 = 63$, so

$$z(MILP) \geq LB = 100 \cdot 63 = 6300.$$

The final comparison

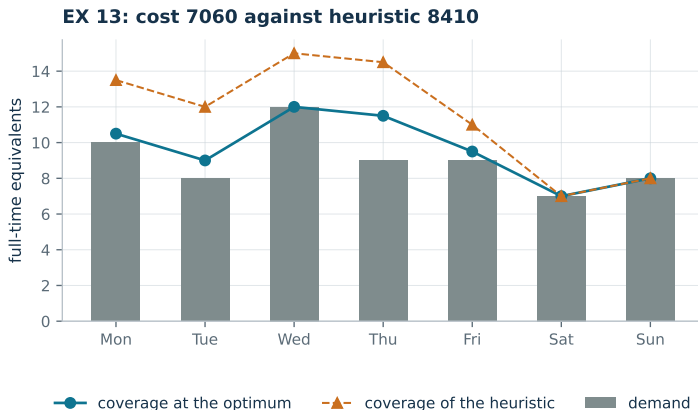
UB	LB	$z(LP)$	$z(LP^+)$	$z(MILP)$
8 410	6 300	115970/17	115970/17	7 060

- ▶ Optimal plan: 5 workers on pattern 1, 4 on pattern 3, 1 on pattern 4, 2 on pattern 5 and 3 on pattern 7, cost 7060
- ▶ Coverage saturated on Wednesday, Saturday and Sunday, in surplus on the other days
- ▶ Heuristic gap: 19.1%
- ▶ $z(MILP) - z(LP) = 7060 - 115970/17 \approx 238$ is the **price of integrality**: people are hired one at a time

Is the part-time day worth it?

- ▶ Alternative contract: four full days and *three* rest days, with no part-time
- ▶ The minimum price per day covered is the same, 100 euros
- ▶ And yet the optimum of the second contract is 6710, against the 7060 of the first
- ▶ The half day falls on a fixed day and often ends up where the coverage is already there: the flexibility lost costs more than the extra coverage
- ▶ “More coverage at the same unit price is always better” is false, and only the model shows it

EX 13: demand and coverage



Daily demand and coverage, in the heuristic solution and in the optimal one.

Summary

- ▶ Set covering with coefficients 1 and $1/2$ and integer variables
- ▶ The costs of the patterns are computed from the data
- ▶ Cost/coverage ratio heuristic: $UB = 8410$
- ▶ Uniform price $t = 100$ on the days: $LB = 6300$
- ▶ Optimum 7060; the price of integrality is about 238
- ▶ The part-time day on a fixed day does worse than rest

EX 14 — Shoes: production, inventory and hirings

Introduction: EX 14

- ▶ A three-month plan: how much to produce, how much to store, how many workers to hire
- ▶ Techniques: **inventory balances** and staff balances
- ▶ Continuous variables for products, integer for people
- ▶ The dual is built by giving every labour hour its true cost

EX 14 — production and workforce over three months

- ▶ Monthly demands: 3000, 5000 and 7000 pairs of shoes
- ▶ At the start: 500 pairs in stock and 100 workers
- ▶ Every worker is paid 1500 euros a month and works 160 hours
- ▶ One pair requires 4 labour hours and 15 euros of raw materials
- ▶ At the start of every month workers may be hired: 100 euros each for training
- ▶ Storing one unsold pair at the end of the month costs 3 euros

The factory wants the production and workforce plan of minimum total cost.

The variables

It is model 9.2A with one more datum, the initial inventory.

x_t = pairs produced in month t , $\forall t \in \{1, 2, 3\}$

s_t = inventory at the end of month t , $\forall t \in \{1, 2\}$

y_t = workers on duty in month t , $\forall t \in \{1, 2, 3\}$

z_t = workers hired in month t , $\forall t \in \{1, 2, 3\}$

- ▶ y_t and z_t count people, so they are integer
- ▶ The demand of the first month is $3000 - 500 = 2500$ net pairs; 14 500 pairs must be produced in total

EX 14 on the instance

EX 14 on the instance

$$\begin{aligned}
 &\min 15x_1 + 15x_2 + 15x_3 + 3s_1 + 3s_2 + 1500y_1 + 1500y_2 + 1500y_3 + 100z_1 + 100z_2 + 100z_3 \\
 \text{subject to} \quad &x_1 - s_1 = 2500 \\
 &x_2 + s_1 - s_2 = 5000 \\
 &x_3 + s_2 = 7000 \\
 &-4x_1 + 160y_1 \geq 0 \\
 &-4x_2 + 160y_2 \geq 0 \\
 &-4x_3 + 160y_3 \geq 0 \\
 &y_1 - z_1 = 100 \\
 &-y_1 + y_2 - z_2 = 0 \\
 &-y_2 + y_3 - z_3 = 0 \\
 &x_1, x_2, x_3 \geq 0 \\
 &s_1, s_2 \geq 0 \\
 &y_1, y_2, y_3 \in \mathbb{Z}_{\geq 0} \\
 &z_1, z_2, z_3 \in \mathbb{Z}_{\geq 0}
 \end{aligned}$$

The LP relaxation and its dual

- ▶ The LP relaxation replaces the integrality of y_t and z_t with non-negativity alone
- ▶ The dual has:
 - ▶ α_t free on the demand balances
 - ▶ $\beta_t \geq 0$ on the hours
 - ▶ γ_t free on the workforce balances
- ▶ The eleven columns of the primal give eleven dual constraints

Dual of EX 14 on the instance

Dual of EX 14 on the instance

$$\begin{array}{llllll}
 \max & 2500\alpha_1 + 5000\alpha_2 + 7000\alpha_3 & & & +100\gamma_1 & \\
 \text{subject to} & \alpha_1 & & -4\beta_1 & & \leq 15 \\
 & & \alpha_2 & & -4\beta_2 & \leq 15 \\
 & & & \alpha_3 & & -4\beta_3 \leq 15 \\
 & -\alpha_1 & +\alpha_2 & & & \leq 3 \\
 & & -\alpha_2 & +\alpha_3 & & \leq 3 \\
 & & & 160\beta_1 & +\gamma_1 - \gamma_2 & \leq 1500 \\
 & & & & 160\beta_2 & +\gamma_2 - \gamma_3 \leq 1500 \\
 & & & & & 160\beta_3 & +\gamma_3 \leq 1500 \\
 & & & & & & -\gamma_1 \leq 100 \\
 & & & & & & -\gamma_2 \leq 100 \\
 & & & & & & -\gamma_3 \leq 100 \\
 & \alpha_1, & \alpha_2, & \alpha_3 & & & \geq 0 \\
 & & & & \beta_1, & \beta_2, & \beta_3 & \geq 0 \\
 & & & & & & & \gamma_1, \gamma_2, \gamma_3 \geq 0
 \end{array}$$

A feasible solution: the primal bound

“Just in time” production: every month the net demand is produced, with no inventory, hiring as many workers as needed.

- ▶ Month 1: 10 000 hours, 63 workers, the 100 already on duty are enough
- ▶ Month 2: 20 000 hours, 125 workers, 25 are hired
- ▶ Month 3: 28 000 hours, 175 workers, another 50 are hired

$$z(MILP) \leq UB = 825\,000.$$

A dual solution: the dual bound

The recipe: $\gamma = 0$, and one hour of work is worth what it really costs,

$$\bar{\beta}_t = \frac{w}{r} = \frac{1500}{160} = \frac{75}{8},$$

so that one pair is worth at most

$$\bar{\alpha}_t = 15 + 4 \cdot \frac{75}{8} = \frac{105}{2} = 52.5.$$

One checks that all the dual columns are satisfied, and

$$LB = \frac{105}{2} \cdot 14\,500 = 761\,250.$$

The final comparison

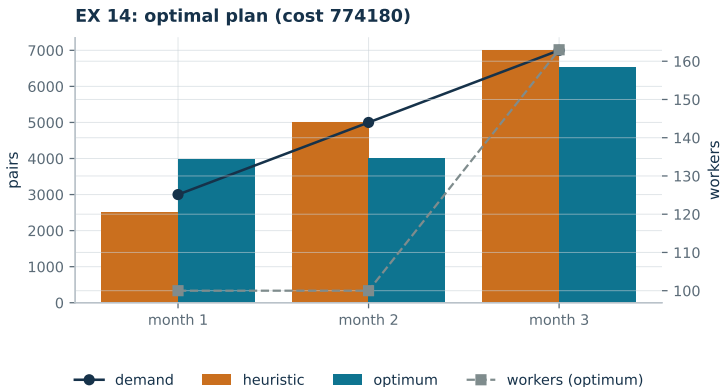
UB	LB	$z(LP)$	$z(LP^+)$	$z(MILP)$
825 000	761 250	773 500	773 500	774 180

- ▶ The optimal plan produces 3980, 4000 and 6520 pairs with 100, 100 and 163 workers
- ▶ 1480 pairs in stock at the end of the first month and 480 at the end of the second
- ▶ 63 workers are hired once only, in the third month

Inventory versus wages

- ▶ Producing early costs 3 euros a pair and saves wages
- ▶ Storage cost at 20 euros: the inventories drop to 1000 and 0 (optimum 807 500)
- ▶ Storage cost at 60 euros: they disappear altogether and one falls back on the heuristic (825 000)

EX 14: the optimal plan



The optimal plan of production, stock and workforce, month by month.

Summary

- ▶ Four families of variables: production, inventory, workforce, hirings
- ▶ Inventory balances, hour constraints, staff balances
- ▶ “Just in time”: $UB = 825\,000$
- ▶ Dual with the hour priced at its true cost: $LB = 761\,250$
- ▶ Optimum 774 180: it pays to produce early and hire once only

EX 15 — The music school timetable

Introduction: EX 15

- ▶ A small weekly timetable with the teachers' preferences
- ▶ Technique: indicators y_{di} linked to the lessons in **both directions** (if and only if)
- ▶ We will see what happens if the link is written one way only
- ▶ And a case in which the dual bound is not enough to prove optimality

EX 15 — weekly timetable

- ▶ Two afternoons (Monday and Tuesday), each of two hours
- ▶ 2 hours of guitar and 2 of violin must be placed
- ▶ Every day must include lessons of at least two different instruments
- ▶ Preferences, to be respected if possible: the guitar teacher would rather not come on Tuesday, the violin teacher would rather not come on Monday
- ▶ The goal is to minimise the number of hours that violate a preference

The variables

Two families of binary variables:

- ▶ $x_{dti} = 1$ if at hour t of day d instrument i is taught
- ▶ $y_{di} = 1$ if instrument i appears on day d

They are eight columns for the lessons — one per triple day, hour, instrument — and four for the indicators.

EX 15 on the instance — primal

EX 15 on the instance — primal

$$\begin{array}{llllll}
 \min & x_{112} & +x_{122}+x_{211} & +x_{221} & & \\
 \text{subject to} & x_{111} & +x_{121} & +x_{211} & +x_{221} & =2 \\
 & x_{112} & +x_{122} & +x_{212} & +x_{222} & =2 \\
 & x_{111}+x_{112} & & & & \leq 1 \\
 & x_{121}+x_{122} & & & & \leq 1 \\
 & x_{211}+x_{212} & & & & \leq 1 \\
 & x_{221}+x_{222} & & & & \leq 1 \\
 & & & & y_{11}+y_{12} & \geq 2 \\
 & & & & y_{21}+y_{22} & \geq 2 \\
 & x_{111} & & & -y_{11} & \leq 0 \\
 & x_{112} & & & -y_{12} & \leq 0 \\
 & x_{121} & & & -y_{11} & \leq 0 \\
 & x_{122} & & & -y_{12} & \leq 0 \\
 & x_{211} & & & -y_{21} & \leq 0 \\
 & x_{212} & & & -y_{22} & \leq 0 \\
 & x_{221} & & & -y_{21} & \leq 0 \\
 & x_{222} & & & -y_{22} & \leq 0 \\
 & -x_{111} & -x_{121} & & +y_{11} & \leq 0 \\
 & -x_{112} & -x_{122} & & +y_{12} & \leq 0 \\
 & & -x_{211} & -x_{221} & +y_{21} & \leq 0 \\
 & & -x_{212} & -x_{222} & +y_{22} & \leq 0 \\
 & x_{111}, x_{112}, x_{121}, x_{122}, x_{211}, x_{212}, x_{221}, x_{222} & & & & \in \{0,1\} \\
 & & & & y_{11}, y_{12}, y_{21}, y_{22} & \in \{0,1\}
 \end{array}$$

Description

- ▶ The first two rows count the hours of each instrument
- ▶ The next four: at most one lesson in each slot of the timetable
- ▶ The two after that are the **variety**: at least two instruments a day
- ▶ Then the two directions of the link between lesson and indicator: eight rows for “if there is a lesson the indicator is on” and four for the converse
- ▶ The hours to place are $2 + 2 = 4$ and the slots $2 \cdot 2 = 4$: every slot holds exactly one lesson

Observation 2: A one-way variety constraint is empty (1/2)

It comes naturally to write only the link $x_{dti} \leq y_{di}$, and not its converse. With that direction alone the constraint $\sum_i y_{di} \geq 2$ is always satisfied: it is enough to set $y_{di} = 1$ without teaching. Solving the model written that way, the solver puts all the guitar on Monday and all the violin on Tuesday — two days with a single instrument each — and declares 0 violated preferences, while the true optimum violates 2.

Observation 3: A one-way variety constraint is empty (2/2)

The wrong model does not merely allow a bad timetable: it also announces a cost that does not exist.

The link $x_{dti} \leq y_{di}$ says “if there is a lesson then the indicator is on”, not the converse. One also needs

$$y_{di} \leq \sum_{t \in T} x_{dti},$$

that is technique 6.10 (if and only if). When a constraint counts indicators, the link must hold in both directions.

A feasible solution: the primal bound

With the corrected model one builds this timetable by hand (the asterisk marks a violated preference):

	hour 1	hour 2
Monday	guitar	violin*
Tuesday	guitar*	violin

- ▶ 2 hours of guitar and 2 of violin, two instruments on each day
- ▶ Every day holds both instruments and disappoints one of them

$$z(MILP) \leq UB = 2.$$

A dual solution: the dual bound

- ▶ All the costs c_{dti} are 0 or 1: the objective is a sum of non-negative terms and $z(MILP) \geq 0$ with no dual needed
- ▶ The dual of the relaxation has one variable per family of primal constraints:
 - ▶ α_i free on the hours of each instrument
 - ▶ $\beta_{dt} \leq 0$ on the slots, $\gamma_d \geq 0$ on the variety
 - ▶ $\delta_{dti} \leq 0$ and $\varepsilon_{di} \leq 0$ on the two directions of the link
- ▶ Its optimum is 0

EX 15 on the instance — dual

EX 15 on the instance — dual

$$\begin{array}{llllllll}
 \max & 2\alpha_1 + 2\alpha_2 + \beta_{11} + \beta_{12} + \beta_{21} + \beta_{22} + 2\gamma_1 + 2\gamma_2 & & & & & & \\
 \text{subject to} & \alpha_1 & +\beta_{11} & & +\delta_{111} & & -\epsilon_{11} & \leq 0 \\
 & & \alpha_2 + \beta_{11} & & +\delta_{112} & & -\epsilon_{12} & \leq 1 \\
 & \alpha_1 & & +\beta_{12} & & +\delta_{121} & -\epsilon_{11} & \leq 0 \\
 & \alpha_2 & & +\beta_{12} & & +\delta_{122} & -\epsilon_{12} & \leq 1 \\
 & \alpha_1 & & & +\beta_{21} & & +\delta_{211} & -\epsilon_{21} \leq 1 \\
 & \alpha_2 & & & +\beta_{21} & & +\delta_{212} & -\epsilon_{22} \leq 0 \\
 & \alpha_1 & & & & +\beta_{22} & +\delta_{221} & -\epsilon_{21} \leq 1 \\
 & \alpha_2 & & & & +\beta_{22} & +\delta_{222} & -\epsilon_{22} \leq 0 \\
 & & \gamma_1 & -\delta_{111} & -\delta_{121} & & +\epsilon_{11} & \leq 0 \\
 & & \gamma_1 & & -\delta_{112} & -\delta_{122} & & +\epsilon_{12} \leq 0 \\
 & & & \gamma_2 & & -\delta_{211} & -\delta_{221} & +\epsilon_{21} \leq 0 \\
 & & & \gamma_2 & & & -\delta_{212} & -\delta_{222} +\epsilon_{22} \leq 0 \\
 & \alpha_1, & \alpha_2 & & & & & \geq 0 \\
 & & \beta_{11}, \beta_{12}, \beta_{21}, \beta_{22} & & & & & \geq 0 \\
 & & & \gamma_1, & \gamma_2 & & & \geq 0 \\
 & & & & \delta_{111}, \delta_{112}, \delta_{121}, \delta_{122}, \delta_{211}, \delta_{212}, \delta_{221}, \delta_{222} & & \leq 0 \\
 & & & & & & \epsilon_{11}, \epsilon_{12}, \epsilon_{21}, \epsilon_{22} & \leq 0
 \end{array}$$

Description of the dual

- ▶ Every row is a column of the primal
- ▶ The first eight are the x_{dti} : the price of the instrument's hour plus that of the slot, corrected by the two directions of the link, cannot exceed the cost of the slot
- ▶ The last four are the y_{di} : switching the indicator on costs the variety and is refunded by the link

The final comparison

$$LB = 0, \quad z(LP) = 0, \quad z(LP^+) = 2, \quad z(MILP) = 2.$$

- ▶ The hand-built timetable was already optimal, but the dual bound does not prove it
- ▶ Between 0 and 2 a gap is left that only branch and bound closes
- ▶ The strengthened relaxation, which adds $x \leq 1$ and $y \leq 1$, reaches 2 instead and settles it

Observation 4: A count that is not a bound on the relaxation

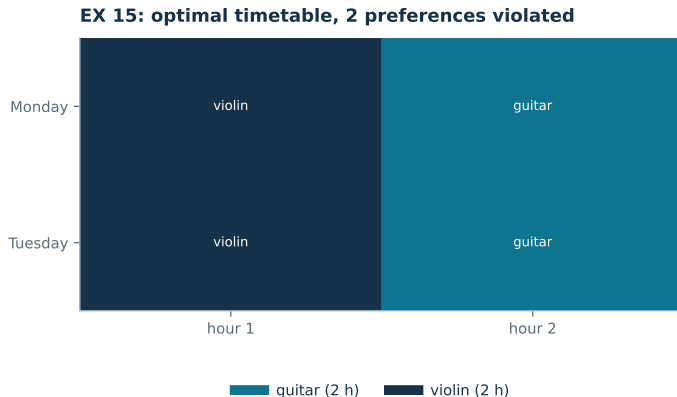
Counting says more: every day holds both instruments, and on each day one of the two teachers would rather not be there; the violations are therefore at least two, and $z(\text{MILP}) \geq 2$.

That count, however, uses the integrality of the variables: on the relaxation it does not hold, and indeed $z(\text{LP}) = 0$. It cannot take the place of LB in the chain of bounds, where every term must hold for the LP as well.

What happens if the preferences get tighter

- ▶ If the guitar teacher would rather not teach at hour 2 of Monday either, the only slot left penalty-free for the guitar is hour 1 of Monday, while the hours to place are 2: at least one more violation
- ▶ If at least *three* different instruments a day are required, three instruments would be needed in only two hours. The model is infeasible, and one proves it by counting: Gurobi returns the status INFEASIBLE, as expected

EX 15: the optimal timetable



The optimal timetable found by the solver, one colour per instrument.

Summary

- ▶ Indicators y_{di} counted by the variety constraint
- ▶ The link must be written in both directions: $x_{dti} \leq y_{di}$ and $y_{di} \leq \sum_t x_{dti}$
- ▶ Hand-built timetable: $UB = 2$; dual bound: $LB = 0$
- ▶ The gap is closed by branch and bound or by the strengthened relaxation
- ▶ A count that uses integrality is not a bound on the relaxation

Final summary (1/2)

The techniques of EX 11–15:

- ▶ EX 11: minimum thresholds and activations with a big-M read off the data
- ▶ EX 12: min-max to translate “balanced”
- ▶ EX 13: set covering with integer, non-binary variables
- ▶ EX 14: inventory and staff balances over several periods
- ▶ EX 15: indicators linked in both directions (if and only if)

Final summary (2/2)

The bounds:

- ▶ In EX 11 and EX 12 the two hand-built bounds coincide: optimality proved without a solver
- ▶ In EX 13 and EX 14 a gap remains, and part of it is the price of integrality
- ▶ In EX 15 the dual bound is 0: branch and bound or a stronger relaxation is needed
- ▶ Every term of the chain of bounds must hold for the LP as well